Historical Vernacular Houses in the Hualien River Basin of Eastern Taiwan: A Spatial Humanities Investigation with Research Data Management Planning

Lim, Ting-iong
61120a009@gms.ndhu.edu.tw
Department of Taiwan and Regional Studies, National Dong Hwa University, Taiwan

Chuang, Tyng-Ruey
trc@iis.sinica.edu.tw
Institute of Information Science, Academia Sinica, Taiwan; Research Center for Information Technology Innovation, Academia Sinica, Taiwan; Research Center for Humanities and Social Sciences (Center for GIS), Academia Sinica, Taiwan

Introduction

Taiwan's rural and suburban vernacular architecture for housing, at least for the period until the end of World War II when Japan ruled the island, can be broadly classified into two main systems: The housing of indigenous peoples and the courtyard-style housing of Han Taiwanese people. The latter type is geographically divided by the Zhuoshui River (濁水溪), which historically shows at least two distinct styles based on the materials they used. In northern Taiwan, the primary construction material was dried clay bricks resulting in a type of housing known as *Thôo-kat-tshù* (literally, soil-brick house; 土埆厝). On the other hand, in the plains and foothills of southern Taiwan including the Zhuoshui River basin, another style called *Tik-kóng-á-tshù* (literally, bamboo house; 竹管仔厝) which utilized bamboo for beams and columns was most prevalent (Aoi et al. 2018). This framework provides a preliminary overview of the spatial distribution of traditional housing styles used by Han Taiwanese in western Taiwan. However, when this framework is applied to examine the historical vernacular houses in eastern Taiwan, a different picture emerges. Specifically, in the Hualien River (花蓮溪) basin of eastern Taiwan, neither *Tik-kóng-á-tshù* nor *Thôo-kat-tshù* is commonly found for old houses in the area. Instead, courtyard-style houses featuring rough clapboard siding, known as Pang-á-tshù, dominate the historical landscape (see Figures 1 and 2).

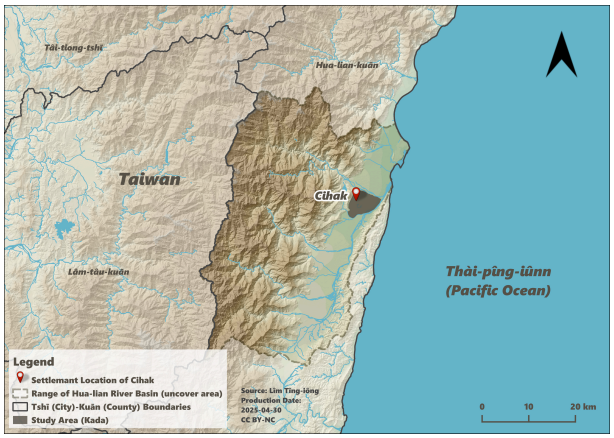

Figure 1. *Pang-á-tshù* in *Cihak* (志學) settlement of the Hualien River Basin.

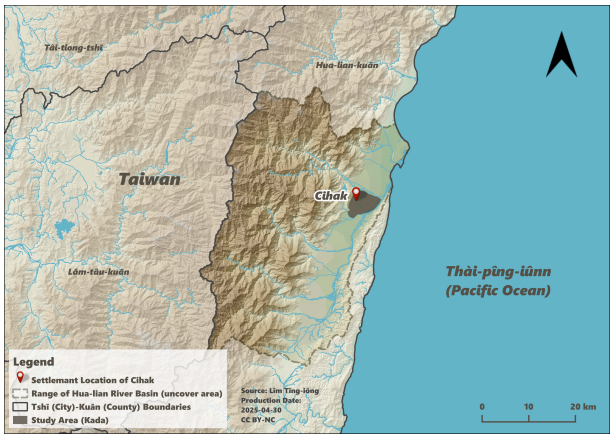

Figure 2. The Location of *Cihak* (志學) in the Hualien River Basin, Taiwan. (The labels are in Taiwanese Romanization, also known as *Tâi-lô* / 台羅.)

Based on preliminary interviews conducted by one of us (Lim) in May 2024 at the *Tāi-hô* (大和) settlement of Hualien County, this unique form of housing has been referred to locally as Pang-á-tshù (literally, clapboard house; 枋仔厝), which means wood plank house in Taiwanese. The facades of this form of housing are covered with highly recognizable (semi-finished) clap-

boards, and the roofs are often covered by Japanese-style cement tiles or metal sheets. At first glance, such a house may remind people of a Japanese-style residence, but the building shows a fairly typical fabric of Han Taiwanese housing. After combing through the settlements and architectural types in the *Tsuí-bué* (水尾) area of Hualien, *Chen* (2022) believes that this kind of courtyard-style residence with clapboard sliding had been influenced by the construction technology of the Japanese colonial period, and he thinks it was the most common feature of Eastern Taiwan buildings for homes. We generally agree with the above assumptions, but the influence of Japanese construction technology alone cannot explain why *Pang-á-tshù* cannot be observed in Western Taiwan. In short, it must be discussed in the context of regional development in Eastern Taiwan considering, for example, factors such as the forestry industries and Japanese immigrant villages that emerged in Hualien during the Japanese colonial period. Our study as reported here is an on-going investigation into the broader socio-economical, geographical, and natural forces that shaped and flourished the *Pang-á-tshù* housing in Eastern Taiwan.

Research Methods

In this study, we investigate the current distribution of historical *Pang-á-tshù* housing in the Hualien River Basin of Eastern Taiwan, with a particular focus on the area historically known as *Kada* (賀田). We combine the perspectives of man-land relationship, regional history, and spatial analysis to understand the production background of this form of housing. Moreover, methods and techniques from Spatial Humanities will serve as the tool sets in our research approach. Humanities such as history and literature have re-introduced theoretical reflections on the concepts of space and place, and finally converged the spatial turn in the humanities (*Flores* & *Martins* 2019). Research works related to spatial humanities in Taiwan mainly started from historical GIS (*Gu* & *Li* 2024), using geographical analysis tools to translate historical maps and archives to more accessible research materials. It is a relatively popular spatial humanities method in Taiwan. Our study is grounded in field work and literature survey, however, but use geospatial methods to manage, analyze, and visualize our data. Considering that the spatial distribution of *Pang-á-tshù* is the core concern of this research, we will collect some basic attribute data of *Pang-á-tshù*, integrate it with the Basic Statistical Areas, and visualize the results through choropleth maps and heatmaps. It is anticipated that this study will provide a preliminary understanding of the spatial distribution of *Pang-á-tshù*, such as their density in the Hualien River Basin. Building upon this foundation, we will further explore the production background of these vernacular housing forms including spatial correlations with logging sites, timber processing factories, and Japanese immigrant villages around the Hualien River Basin. Through these analyses, we aims to reveal the overall picture of vernacular housing in Eastern Taiwan, serving as a basis for comparative studies with vernacular housing in other regions of Taiwan.

Data Management Planning

A crucial part of this study is the management of research data during our investigation process. Previously we have used a research data repository, the depositar [0], as a central hub for the curation and presentation of the datasets and media files collected for a summer internship project on documenting a religious festival held in a temple in *Báng-kah* (艋舺), Taipei. By utilizing the Jupyter Book toolkit [1], we authored a collection of web documents by the end of the project. Please refer to the project dataset [2] and the online project webpages [3]. For the current investigation, we plan to continue this practice to disseminate our materials and findings, and to make them widely accessible. Furthermore, we will start to develop a Data Management Plan (DMP) for the current project with the goal of practicing the FAIR Data Principles [4]. The *depositar*, the data repository we use, has been developed for the use of worldwide researchers by a team of researchers at Academia Sinica in Taiwan. It is an open repository meeting the FAIR Data Principles [5].

[0] https://data.depositar.io/about
[1] https://jupyterbook.org/
[2] https://pid.depositar.io/ark:37281/k5h5s0w5r
[3] https://iong2k.github.io/senior_project/
[4] https://force11.org/info/the-fair-data-principles/
[5] https://pid.depositar.io/ark:37281/k5n4c2v0n

Bibliography

Aoi, A. / Chang, T. / Yang, C. (2018): “City of Bamboo – Inland Port Towns in Zhuoshui River Basin and Historical Significance of Bamboo Town Houses (In Traditional Chinese)”, in: Journal of Architecture 104: 103–121. < https://doi.org/10.3966/101632122018060104007 >

Chen, Y. (2022): “Spatial Fabric and Land Structure of the Old Shuiwei Settlement in Ruisui (In Traditional Chinese)”, in: Huang, S. (ed.): Ethnic Groups, Industries, and Local Society in the Xiuguluan River Basin. Taipei: SMC Publishing: 245–277. < http://ir.sinica.edu.tw/handle/201000000A/84305 >

Chuang, T.-R. / Ho, M.-S. / Lee, C.-J. / Wang, C.-H. (2024): “Building An Open Repository for FAIR Data: Experience Sharing on Technologies, Communities, and Policies”, in: Open Repositories 2024 (OR2024), June 28, Gothenburg, Sweden. Zenodo. < https://doi.org/10.5281/zenodo.12579625 >

Ku, Y. / Li, C. (2024): “Historical GIS in Taiwan: Retrospect and Prospect (1990–2020) (In Traditional Chinese)”, in: Journal of Digital Archives and Digital Humanities 13: 1–41. < https://doi.org/10.6853/DADH.202404_(13).0001 >

Murrieta-Flores, P. / Martins, B. (2019): “The Geospatial Humanities: Past, Present and Future”, in: International Journal of Geographical Information Science 33(12): 2424–2429. < https://doi.org/10.1080/13658816.2019.1645336 >